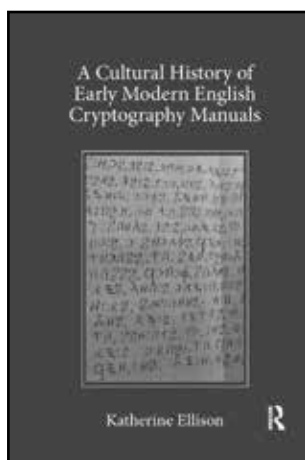
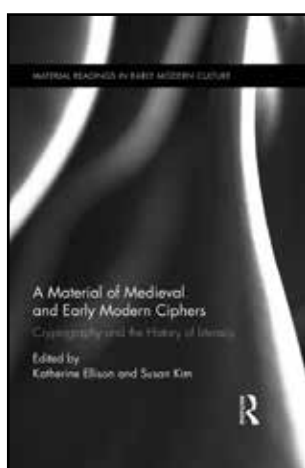


BOOK & FILM REVIEWS



Katherine Ellison, Susan Kim (eds.), *A Material History of Medieval and Early Modern Ciphers. Cryptography and the History of Literacy*

(Oxford-New York: Routledge 2018), 286 pages, ISBN: 978-1-138244641, eBook GBP 44,99; Hardback GBP 115,00.



Katherine Ellison, *A Cultural History of Early Modern English Cryptography Manuals*

(Oxford-New York: Routledge 2017), 218 pages, ISBN: 978-0-367139339, Paperback and eBook GBP 38,99; Hardback GBP 125,00.

This review covers two books by Katherine Ellison: her 2017 monograph *A Cultural History of Early Modern English Cryptography Manuals* as well as the co-edited volume (together with Susan Kim) *A Material History of Medieval and Early Modern Ciphers. Cryptography and the History of Literacy*, published in 2018. Both books deal with the subject of less well studied aspects in the history of cryptology.

A Material History of Medieval and Early Modern Ciphers is an edited collection bringing together interdisciplinary scholars from fields such as history, mathematics and the arts. Historical ciphering practices are studied from a variety of theoretical perspectives such as new materialist, media studies or cognitive studies. Sources on ciphering are seen as documents of reading and book history but also of the culture of orality at the period concerned. Ciphering is approached as an occult art, an aesthetic but also as a mathematical language. The analyses focus on cultural contexts of ciphering, such as the

rhetorical situations at play and the role of ciphering related to literacy. The focus is on book history and the significance of the materiality of text for cultural studies. It is not just a showcase of ciphers solved like it is common with many cryptological publications.

The introduction provides a brief historical overview of ciphering in the Western world, starting from runic ciphers and leading up to the early modern period. Under the heading of “Ciphers and Literacy”, basic concepts of cryptology are explained. That way, the book is fit to read for people without an explicit background in cryptology. The approach of investigating reading practices and literacy via the history of cryptography is an interesting one without doubt which has led to a formidable collection of essays.

In the early modern cypher age, a sort of ciphering education started to become part of popular culture, which not only massively increased the citizens’ cipher literacy but even made considerable amounts of individuals intuitive cryptanalysts. “Ciphered communication, whether in etched stone and bone, in musical notae, runic symbols, polyalphabetic substitution, algebraic equations, graphic typographies, or literary metaphors, took place in contested social spaces and offered a means of expression during times of political, economic, and personal upheaval,” the book cover stresses. The collection covers a plethora of media and is a very valuable resource for the history of ciphering. Unlike many books in the history of cryptology, it not only represents a collection of ciphers solved but also contextualizes them culturally. It does not assume previous knowledge of neither cryptology nor its mathematical aspects. The edited collection features 11 relatively short articles plus an introduction and afterword by the editors. Apart from Quinn Dupont’s “The Printing Press and Cryptography: Alberti and the Dawn of a Notational Epoch” and Karen Britland’s “‘What I write I do not see’: Reading and Writing With Invisible Ink”, however, the material culture emphasized in the book title does not seem

very dominant apart from the fact that ciphering is generally a more material-related activity than mere reading. Also, the subtitle indicating a focus on the history of literacy might be considered somewhat misleading because this subject does not overall seem to be a major point of focus in the collection itself. It is the breadth of media and contexts covered which should be considered the greatest asset of the collection. All in all, the book makes for a very important resource in the historiography of medieval and early modern cryptology.

Ellison's monograph *A Cultural History of Early Modern English Cryptography Manuals*, on the other hand, deals with 17th century English language cryptography manuals, notably John Wilkins' *Mercury; or, The Secret and Swift Messenger* (1641), Noah Bridge's *Steganographie and Cryptographie* (1659) and its sequel *Rarities: or the Incomparable Curiosities in Secret Writing* (1665, published anonymously), Samuel Morland's *New Method of Cryptography* (1666) and John Falconer's *Cryptomenysis Patefacta: Or the Art of Secret Communication Disclosed without a Key* (1685). Ellison argues that the boom of this genre marks a turning point in intellectual and reading history, yet it had previously been disregarded by cryptology scholarship because the ciphers from didactical manuals are not necessarily exciting or new in and of themselves. Ellison seeks out to show what can be learned from such texts. However, it has to be noted that the above-mentioned treatises related to the English Civil War are not all manuals and also not solely cryptographical, as the book title seems to indicate.

Ellison suggests the fact that a didactical genre became established shows that ciphering became to be seen as fashionable, that reading practices were changed in favor of the engagement with material aspects of text like it has to be done with ciphers and also that ciphering became seen as an ability demonstrating intellect. Thus, a skill crucial to intelligence (in the Intelligence Studies sense) became part of the lives of a large group of citizens in the turbulent time of the 17th century. The popularization of ciphering not only created cryptological experts: early exposure to this kind of didactic material also made sure its recipients would develop intuitive capacities for ciphering and deciphering, something Ellison refers to as cipher literacy. Ellison points out that 17th century cryptography manuals illustrate the early capitalization of information. She set out

to disperse the long-held opinion of the alleged naïveté of 17th century ciphers and the cluelessness of cryptological writings by contextualizing them in their cultural context, thus making their intentions easier to understand for a modern reader.

Ellison addresses an aspect yet unaddressed in the history of cryptography, filling a gap in an area of research which had thus far been overlooked. However, large parts of the book are devoted to the cultural links between cryptography and the history of reading as well as the novel which are of little to no avail for the historiography of cryptography. The books of secrets tradition, an incredibly popular genre at the time and an important point for comparison, as well as the dynamics of its rhetoric claiming the revelation of secrets might have been investigated in more detail, especially given the genre studies approach of the book.

The first chapter "The structures of seventeenth-century cryptography manuals" investigates how authors narrate the historical foundations of cryptography in the introductions to their works, establishing cryptology as a long-standing tradition. Furthermore, the details of such retellings matter: they allow us to gather an understanding of which parts of the cryptological tradition were valued by the 17th century authors. Ellison stresses, for example, that Wilkins seems to have been influenced strongly by the new rhetoric around the scientific experiment, paving the way for the establishment of cryptology as a scientific endeavor rather than the occult practice it had been perceived as beforehand. Chapter 2 "Textual precedents and seventeenth-century cryptography manuals" gives a useful overview on the tradition of the textual transmission of cryptography, yet it is unfortunate that contemporary Latin sources were not considered for the early modern period. Additionally, the background on contemporary Latin manuals is somewhat lacking as well: Athanasius Kircher does not seem to be mentioned at all and his student Kaspar Schott is misspelled as "Gasparis Schotti" (the genitive form of his Latinized name).

Chapter 3 "Occult communication, the new sciences, and cryptography" illustrates Wilkins's attempts to free cryptology from its association with the occult. Later chapters investigate the universal language discussion with its relation to cryptology and trade intelligence as capital in early modern commerce. Despite the fact that the related early modern books of secrets tradition is mentioned in passing,

unfortunately, Jütte's highly relevant concept of the "economy of secrets" is missing from the discussion (Daniel Jütte, *The Age of Secrecy: Jews, Christians, and the Economy of Secrets, 1400-1800*; New Haven 2015, an English translation of a German PhD thesis). Lastly, Ellison illuminates the relationship between the idea of intelligence and methods for solving cryptography (such as frequency analysis) as well as the materiality and multimodal communication strategies of cryptography manuals.

A Cultural History of Early Modern English Cryptography Manuals represents a genre studies approach, which is somewhat unusual in the historiography of cryptography. Most books about cryptography list ciphers and their solutions, often disregarding the cultural context of those ciphers. Here, Ellison fills a considerable gap, not only by

looking at a time which had previously been neglected but also by studying the conventions and contexts of the genre in and of itself. The book achieves this by investigating five texts in the form of close readings, however, Johan Wilkins's *Mercury* was given the most space by far. Summing up, the book fills a blank in the historiography of cryptology and illuminates cultural history aspects which have thus far been understudied. Those interested in the history of cryptology should ask their libraries to purchase not only Ellison's monograph but also the edited collection. While they both cover historical aspects of cryptology, their focus is not identical. In combination, they make a great addition to any library on historical cryptology, covering a wide range of topics the cryptologist research community had somewhat neglected thus far.

Reviewed by Sarah Lang



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MMag. phil., MA MA, studied Classics, History, Philosophy and Religious Studies in Graz and Montpellier before working as a Digital Humanities scholar at the Graz Centre for Information Modelling in 2016. In her PhD thesis, she developed a tool for the digital analysis of the alchemical cryptographical and stylistic device called Decknamen, using the Neo-Latin corpus of iatrochymist Michael Maier (1568-1622). Her research interests include the early modern history of science and its relation to secrecy and cryptography, alchemy and metallurgy, as well as Classics using digital methods.

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